

Valence Alignment Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Contrasts on the Exchange-Aligned Data Model

PENPAL Project

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This notebook compares **valence alignment** across the three experimental conditions using the new exchange-aligned structure.

- **Human-AI**: Human co-writing with an LLM
- **Human-Human**: Two humans co-writing together
- **AI-AI**: Two AI agents co-writing together

Important structural note:

- `author_1` / `author_2` are now treated as **slot 1** / **slot 2** in the cleaned exchange sequence.
- In **Human-AI**, those slots also map onto human vs AI roles.
- In **Human-Human** and **AI-AI**, the slots are positional sides of the exchange, not stable identity labels.

The analysis window is restricted to **complete exchanges** and the default aligned turn range (`analysis_turn` 1 to 9).

Primary cross-condition object. Each story contributes one unsigned, label-invariant dyad asymmetry magnitude:

$$A_{\text{valence}}(i) = |\text{atanh}(r_{1 \rightarrow 2, i}) - \text{atanh}(r_{2 \rightarrow 1, i})|$$

where $r_{1 \rightarrow 2, i}$ is the per-story Pearson correlation between same-turn slot 1 and slot 2 valence, and $r_{2 \rightarrow 1, i}$ is the correlation between slot 2's previous-turn valence and slot 1's current-turn valence. The main question

is whether A_{valence} differs across conditions. The directional-slope LMMs are retained as supplemental views. Starter-mechanism and Human-AI speaker-type decompositions are deferred to a follow-up pass.

Article-use note. The primary cross-condition claim should use the unsigned per-story A_{valence} test below. Signed directional slot models are diagnostic only: in Human-Human and AI-AI, Slot 1 / Slot 2 are positional labels rather than theoretically meaningful agent identities.

1 Load Data

Data availability:

A tibble: 6 x 4

## file	`human-ai`	`human-human`	`ai-ai`
## <chr>	<lgl>	<lgl>	<lgl>
## 1 dyadic_sentiment_scores.parquet	TRUE	TRUE	TRUE
## 2 interaction_level_surface_metrics.parquet	TRUE	TRUE	TRUE
## 3 novelty_scores.csv	TRUE	TRUE	TRUE
## 4 semantic_exploration_binned.parquet	TRUE	TRUE	TRUE
## 5 story_embeddings_interaction_level.parquet	TRUE	TRUE	TRUE
## 6 interaction_level_stories_filtered.csv	TRUE	TRUE	TRUE

Table 1: Loaded exchange-aligned valence data by condition

condition	n_stories	n_turns	min_turn	max_turn
Human-AI	100	851	1	9
Human-Human	36	324	1	9
AI-AI	40	360	1	9

2 Preprocessing

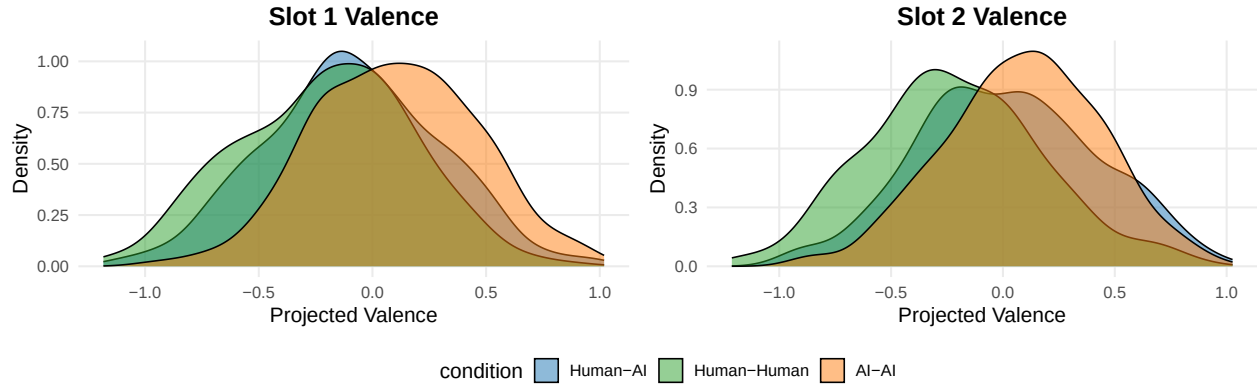
Analysis rows: 2718

Stories: 176

3 Descriptive Checks

Table 2: Mean projected valence by condition

condition	n_stories	n_turns	slot_1_mean	slot_2_mean
Human-AI	100	751	-0.091	0.018
Human-Human	36	288	-0.205	-0.218
AI-AI	40	320	0.100	0.090



4 Primary: Per-Story Asymmetry Magnitude

Stories with both directional correlations available:

Table 3: Per-story A_valence sample sizes by condition

condition	n
Human-AI	100
Human-Human	36
AI-AI	40

Table 4: Per-story |Fisher-z asymmetry| by condition

condition	n	mean_A	sd_A	median_A
Human-AI	100	0.568	0.472	0.447
Human-Human	36	0.463	0.375	0.381
AI-AI	40	0.482	0.403	0.359

One-way ANOVA: A_valence ~ condition

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## condition    2    0.4  0.1978   1.028   0.36
## Residuals  173   33.3  0.1925
```

##

Kruskal-Wallis (robust): A_valence ~ condition

##

Kruskal-Wallis rank sum test

##

data: A_valence by condition

Kruskal-Wallis chi-squared = 1.8801, df = 2, p-value = 0.3906

##

Pairwise Wilcoxon (BH-adjusted):

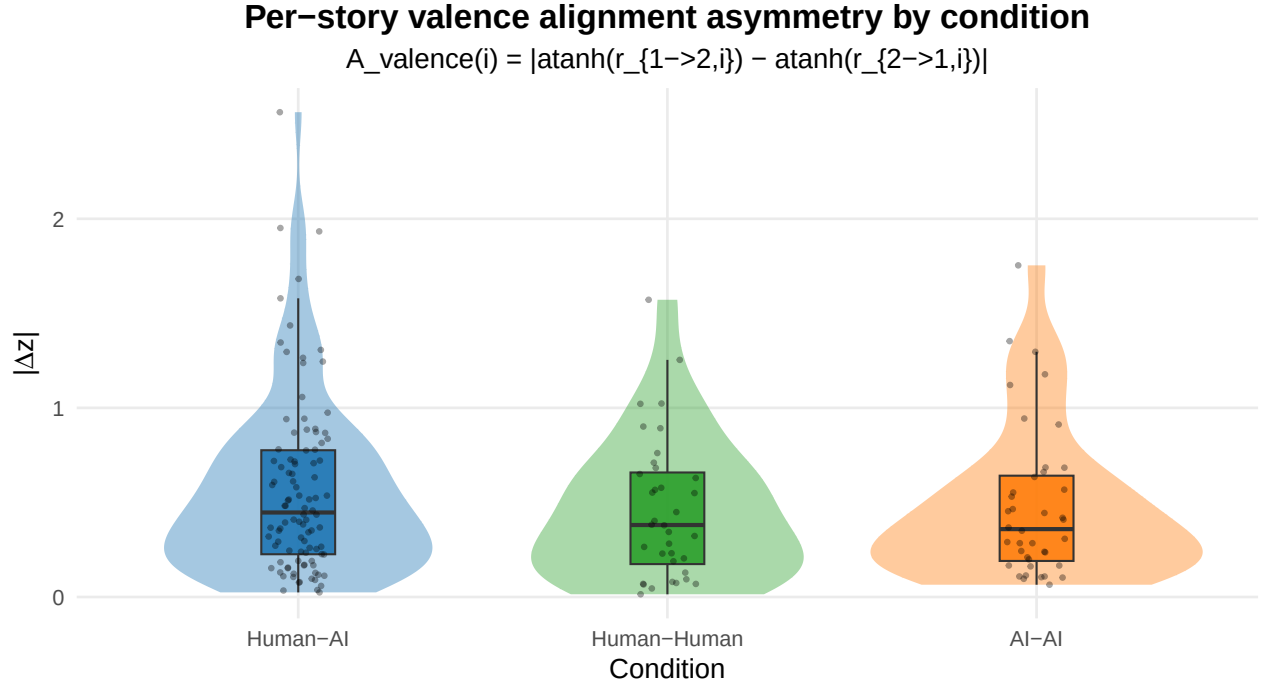
##

Pairwise comparisons using Wilcoxon rank sum test with continuity correction

##

data: df_asym\$A_valence and df_asym\$condition

```
##
##           Human-AI Human-Human
## Human-Human 0.44      -
## AI-AI        0.44      0.78
##
## P value adjustment method: BH
```



5 Bootstrap CIs and Effect Sizes on $A_valence$

Table 5: Bootstrap 95% CI on per-condition mean of $A_valence$ ($R = 10,000$)

condition	n	mean	ci_lower	ci_upper
Human-AI	100	0.568	0.481	0.663
Human-Human	36	0.463	0.347	0.587
AI-AI	40	0.482	0.366	0.613

Table 6: Pairwise Cliff's delta on $A_valence$

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	100	36	0.127	negligible
Human-AI vs AI-AI	100	40	0.114	negligible
Human-Human vs AI-AI	36	40	-0.039	negligible

6 Planned Contrast: HA vs $(HH + AA)/2$

Table 7: Planned contrast: A_valence in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	0.095	0.065	1.469	171.523	0.144	-0.033	0.224

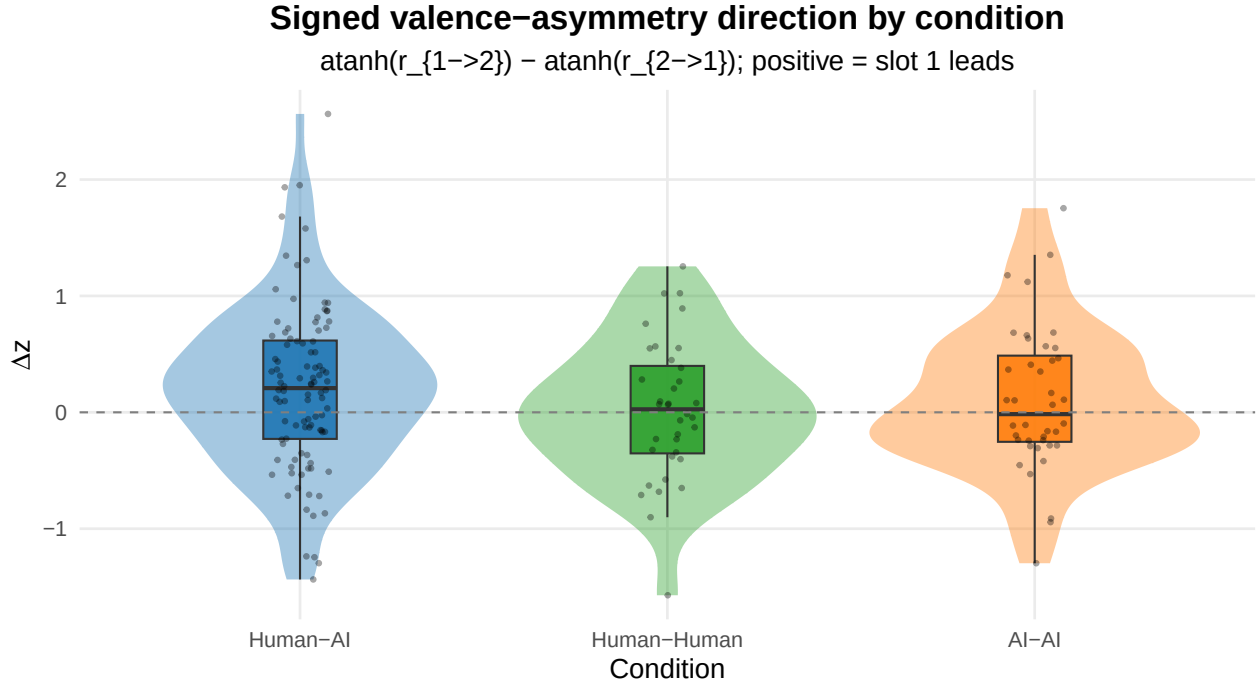
7 Directional Stability of Signed Valence Asymmetry

Tests whether the *direction* of the slot-1 vs slot-2 alignment difference is consistent across stories within each condition. The signed quantity is $\text{signed}_i = \text{atanh}(r_{1 \rightarrow 2, i}) - \text{atanh}(r_{2 \rightarrow 1, i})$. HA-specific identity-tied directionality predicts high **stability** and **sign_consistency** in HA, low in HH and AA, even when unsigned A_valence is similar.

Table 8: Per-condition directional stability of signed valence asymmetry

condition	n	mean	sd	stability	sign_consistency	one_sample_t_p
Human-AI	100	0.203	0.712	0.285	0.61	0.005
Human-Human	36	0.014	0.601	0.023	0.50	0.889
AI-AI	40	0.107	0.623	0.172	0.50	0.285

Brown-Forsythe variance-equality p = 0.5982



8 Permutation Test on A_valence ~ condition

Table 9: Permutation test on $A_{\text{valence}} \sim \text{condition}$ ($R = 10,000$ story-label permutations)

F_obs	p_perm	R
1.0276	0.3608	10000

9 Summary

Primary cross-condition result: per-story $A_{\text{valence}} = |\text{atanh}(r_{1 \rightarrow 2, i}) - \text{atanh}(r_{2 \rightarrow 1, i})|$, tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of A_{valence}
- pairwise Cliff’s δ effect sizes
- planned single-df contrast HA vs (HH+AA)/2
- 10k story-label permutation test on $A_{\text{valence}} \sim \text{condition}$

The Human-AI signed Human/AI directional decomposition lives in `analysis/human-ai/valence_alignment_analysis.Rmd` and is not duplicated here.

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)

Analysis completed: 2026-05-01 23:19:51.59371